

Designing a Mixed Reality tool to previsualize shooting scenarios.

An end-to-end design process for a spatial computing application that enables production teams to create, adjust, and share MR shooting scenarios — before a single crew member sets foot on location.

UX Research UX Design Design System Prototyping

THE PROBLEM

Previsualizing shooting scenarios was costly, slow and imprecise.

Production teams had been relying on a patchwork of manual processes — hand-drawn floor plans, printed storyboards, physical walkthroughs and time-consuming 3D renders — to agree on how a shoot should look. None of these methods could show everyone the same picture, and misalignments were only discovered on location, when fixing them was most expensive.

GOAL

Design an intuitive Mixed Reality tool that lets directors, producers, and set designers collaboratively design, adjust, and evaluate shooting scenarios — in real time, from anywhere.

- **Too much time**
Pre-production planning cycles stretched across weeks due to iterative, back-and-forth floor plan revisions and on-site walkthroughs.
- **High costs**
Physical test setups, travel to locations, and emergency re-shoots to fix misunderstood framing decisions drained budgets before filming started.
- **Lack of precision**
2D floor plans and verbal briefs left too much room for interpretation. Camera angles, lens choices, and blocking rarely matched the creative vision.
- **Complex communication**
Directors, DPs, and producers struggled to share a common spatial reference — descriptions like 'wider shot' meant different things to different people.

RESEARCH

Research focused on real production workflows.

Before sketching a single wireframe, we embedded ourselves in the industry. Eleven weeks of contextual research across four production companies revealed how decisions are actually made — and where things break down.



In-depth Interviews

8 interviews with directors, executive producers, directors of photography, and set designers across narrative film and commercial shoots.



On-site Observation

3 full shooting days observed, shadowing crew through setup, blocking, and lighting decisions to map real-world friction points.



Workflow Analysis

Mapped end-to-end pre-production flows across four studios — from creative brief to shooting schedule — to identify handoff breakdowns.



Early Concept Testing

Low-fidelity paper prototype sessions with 6 participants to stress-test spatial interaction assumptions before committing to digital form.

KEY INSIGHTS

- ✓ 100% of participants had experienced on-set surprises that could have been avoided with better previsualization.
- ✓ Directors spend an average of 3.2 hours per shooting day resolving spatial misunderstandings.
- ✓ Remote collaborators (producers, exec clients) felt most excluded from creative decisions.
- ✓ Existing 3D tools require specialist skills — non-technical stakeholders disengage entirely.
- ✓ The moment a spatial reference is shared visually, decision confidence increases dramatically.

PRIMARY PERSONA

Lucia M.
Art Director — 32 years
8 years experience

Perfectionist Visual thinker

Time-pressed

"I need to see how everything will actually look on camera before I move the entire crew to location. There's no way to explain that with a floor plan."

Goal: Align team on visual language early
Frustration: Tools require specialist training
Device: HoloLens 2, iPad Pro

DESIGN

From information to experience.

A four-phase design process grounded in research synthesis, divergent exploration, component-first execution, and collaborative validation with real production teams.

01

Define

02

Ideate

03

Design

04

Validate

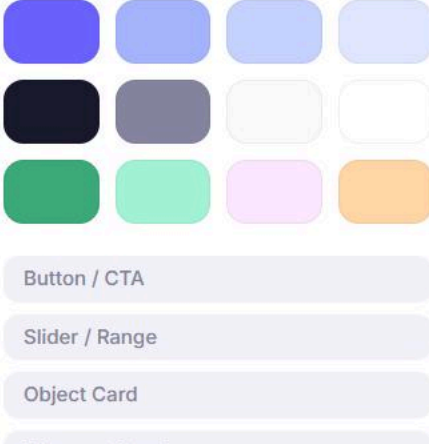
INFORMATION ARCHITECTURE

Hierarchical structure distilled from workflow mapping sessions with three studios.



COMPONENT SYSTEM

Spatial UI kit — 48 components optimized for hand-tracking and gaze input.



MR INTERACTION PATTERNS

Hand-tracking gestures validated across HoloLens 2 and Magic Leap 2 hardware.

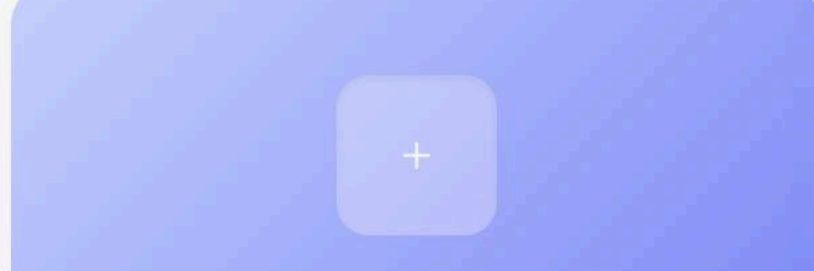


All gestures tested with 12 participants over 4 rounds of usability sessions, iterating on grab tolerances and collision radii.

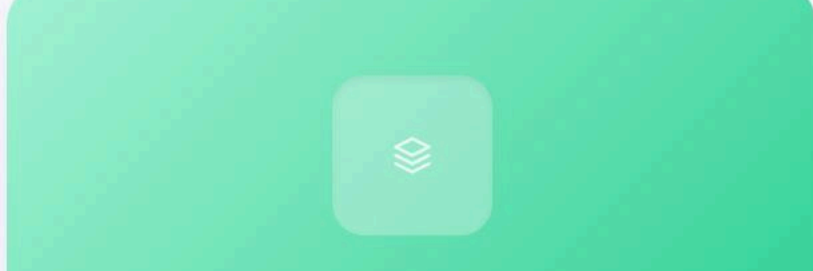
RESULT

An MR app to design scenarios with confidence and speed.

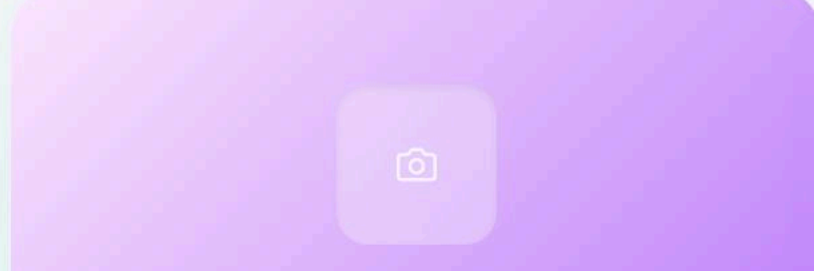
The final product is a Mixed Reality application that runs natively on head-mounted devices and mirrors state to a companion iPad app — so producers and remote stakeholders can follow along in real time without wearing a headset.



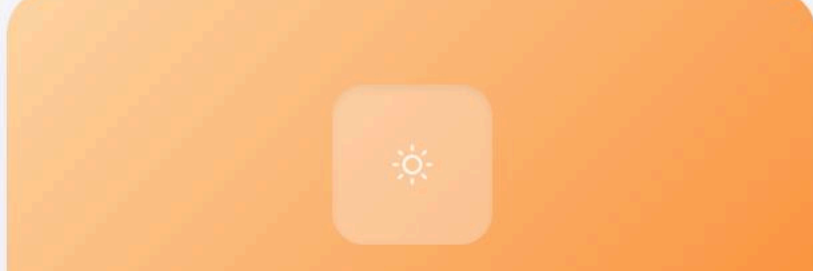
01 — Create a Scenario



02 — Add Elements



03 — Cameras & Framing



04 — Lighting & Atmosphere

IMPACT

Fewer assumptions, more confident decisions.

Measured across a 3-month pilot with two production companies and validated through post-shoot retrospectives, stakeholder interviews, and time-tracking data.



-60%

Reduction in pre-production planning time — from an average of 14 days to 5.5 days per project.



-30%

Cost reduction in test setups and on-location adjustments across the pilot cohort.



94%

Of participants reported making more accurate creative decisions compared to their previous workflow.



×3.2

Increase in remote stakeholder engagement during pre-production — from observer to active participation.

TAKEAWAYS

Key takeaways

- Spatial context is the missing layer in production communication.**
Every tool in the production stack communicates in 2D. The moment you add a shared 3D reference, alignment happens faster and with less friction.
- Simplicity is harder to design in XR than in flat interfaces.**
Hand-tracking UX required ruthless reduction. Every interaction that felt 'obvious' on a screen became a cognitive load challenge in a headset.
- Design for the skeptic in the room.**
The most valuable design decisions came from accommodating the most tech-averse person on the team — the veteran AD who had never used VR.
- Remote participation is a first-class use case, not an afterthought.**
The companion iPad app was scoped as a stretch goal but became the primary access point for producers and clients during pilots.

NEXT STEPS

What comes next



Real-time Collaboration

Multi-user sessions where director, DP, and producer can annotate the same spatial scene simultaneously from different devices.



Asset Library

An integrated catalog of set pieces, prop dimensions, and crew equipment — searchable, scalable, and contributed by the community.



Plan & Report Export

One-tap export of camera sheets, floor plans, and lighting diagrams directly from the MR session to PDF and production management tools.